RoutePilot

Last-Mile Delivery Optimizer

A simple platform to manage delivery orders, drivers and more efficient delivery routes.

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Problem Statement

Last-mile delivery operations can become difficult to coordinate as orders increase.

01

Manual coordination

Managing multiple orders and drivers manually can take time and effort.

02

Inefficient routes

Choosing delivery sequences without route support can lead to unnecessary travel.

03

Limited visibility

Order status, driver capacity and delivery progress may be spread across different records.

Need: one simple place to organize delivery information and support route planning.

Proposed Solution

RoutePilot brings key delivery operations into one lightweight dashboard.

ROUTEPILOT

Manage orders.
View drivers.
Plan better routes.

WHAT THE PLATFORM DOES

01 **Centralizes delivery orders**

Create orders and update delivery status.

02 **Shows driver information**

View driver capacity and current load.

03 **Suggests a delivery sequence**

Uses a nearest-neighbour heuristic for pending stops.

04 **Summarizes operations**

Displays order counts and delivery activity.

System Architecture

A simple three-layer flow connects the interface to application logic and stored data.

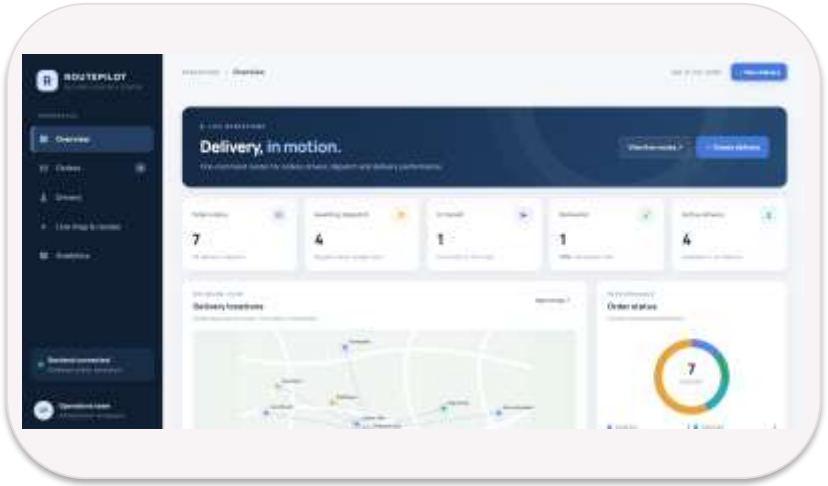


User action → API request → backend processing → database read/write → response on dashboard

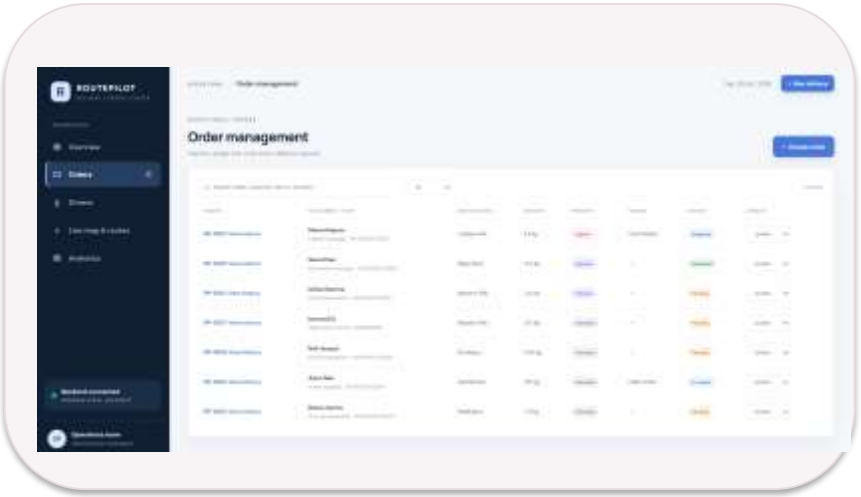
Route planning uses a nearest-neighbour heuristic and an illustrative distance matrix.

RoutePilot Interface & Module Demonstration

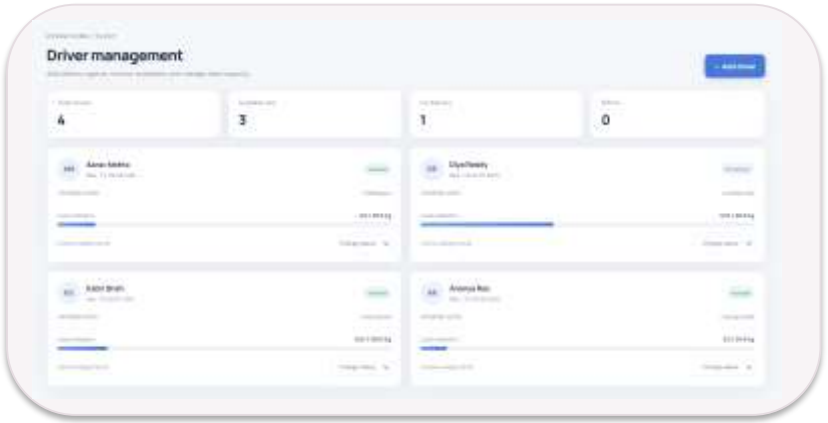
Dashboard



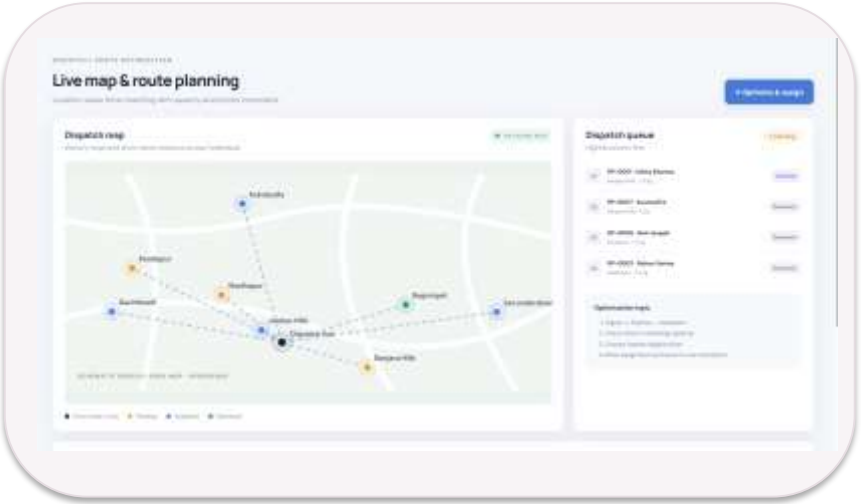
Order Management



Driver Management



Route Optimization & Analytics



Technology Stack & Project Scope

Technologies used in the current lightweight local version.

01

Frontend

HTML, CSS, JavaScript

02

Backend

Python, FastAPI, Uvicorn

03

Database

SQLite

04

Route logic

Nearest-neighbour heuristic

Project scope: a classroom prototype demonstrating selected concepts across CO1–CO6.

THANK YOU

A clearer way to coordinate deliveries.